Ex. 6 – Cryptography & Steganography

6.a: Classical Ciphers: Implement Caesar Cipher and Vigenère Cipher using a Linux shell script, perform encryption and decryption on sample text files, verify correctness by comparing plaintext and recovered text.

1. Implement Caesar Cipher using a Linux shell script,

Create file caesar.sh in vi editor: vi caesar.sh

Press 'i' to enter into insert mode

Shell Script : Encryption & Decryption using Caesar Cipher Method)

```
mode=$1
shift=$2
input=$3
output=$4
 # Create Caesar mappings
low_src=abcdefghijklmnopqrstuvwxyz
low_dst=$(echo "$low_src" | cut -c$((shift+1))-)$(echo "$low_src" | cut -c1-
$shift)
echo $low_dst
up_src=ABCDEFGHIJKLMNOPQRSTUVWXYZ
up_dst=$(echo "$up_src" | cut -c$((shift+1))-)$(echo "$up_src" | cut -c1-$shift)
echo $up_dst
if [ "$mode" = "enc" ]; then
   tr "$low_src" "$low_dst" < "$input" |
   tr "$up_src"  "$up_dst"  > "$output"
echo "Encrypted Text="
cat "$output"
elif [ "$mode" = "dec" ]; then
   tr "$low_dst" "$low_src" < "$input" |
   tr "$up_dst"  "$up_src"  > "$output"
echo "Decrypted text="
cat "$output"
else
   echo "Usage: $0 enc|dec shift infile outfile"
fi
```

Execution & Sample Output:

1.Create a sample file:

```
# cat > plain.txt <<EOF
> Hello World
> EOF
```

2. Make executable:

```
chmod  +x  caesar.sh
```

3. Caesar Cipher Execution & Verification:

```
# ./caesar.sh enc 3  plain.txt cipher.txt
defghijklmnopqrstuvwxyzabc
DEFGHIJKLMNOPQRSTUVWXYZABC
Encrypted Text=
Khoor Zruog
#./caesar.sh dec 3  cipher.txt recover.txt
defghijklmnopqrstuvwxyzabc
DEFGHIJKLMNOPQRSTUVWXYZABC
Decrypted text=
Hello World
#diff plain.txt recover.txt
```

No output from diff ⇒ **encryption and decryption are correct**

2. Implement Vigenère Cipher using a Linux shell script, perform encryption and decryption on sample text files, verify correctness by comparing plaintext and recovered text.

Algorithm:

The Vigenère cipher uses a keyword to determine a sequence of shifts.

For each letter:

- Convert plaintext letter → number (A=0 … Z=25)
- Convert key letter → number
- Encryption:

  $C=(P+K) \bmod 26$
- Decryption:

  $P=(C-K+26) \bmod 26$

Non-alphabetic characters (spaces, punctuation) are left unchanged.

Shell Script : Encryption & Decryption using Vigenère Cipher Method)

```
mode=$1
key=$2
input=$3
output=$4
if [ $# -ne 4 ]; then
    echo "Usage: $0 enc|dec KEY input output"
    exit 1
fi
# Convert key to uppercase and remove non-letters
key=$(echo "$key" | tr 'a-z' 'A-Z' | tr -cd 'A-Z')
keylen=${#key}
awk -v mode="$mode" -v key="$key" -v keylen="$keylen" '
BEGIN{FS="";j=0}
{
for(i=1;i<=NF;i++){
c=$i
if(c~/[A-Za-z]/){
p=index("ABCDEFGHIJKLMNOPQRSTUVWXYZ",toupper(c))-1
```

```
k=index("ABCDEFGHIJKLMNOPQRSTUVWXYZ",substr(K,j+1,1))-1
v=(mode=="enc")?(p+k)%26:(p-k+26)%26
o=substr("ABCDEFGHIJKLMNOPQRSTUVWXYZ",v+1,1)
printf(c~/[a-z]/)?tolower(o):o
j=(j+1)%keylen
}else printf c
}}
END{printf"\n"}
' "$input" > "$output"
echo "Output written to: $output"
```

Execution & Sample Output:

1.Create a sample file:

```
# cat > plain.txt <<EOF
> hello world
> EOF
```

2. Make executable:

```
chmod  +x  vigenere.sh
```

3. Vigenere Cipher Execution & Verification:

```
# ./vigenere.sh enc KEY  plain.txt cipher.txt
Output written to: cipher.txt
# ./vigenere.sh dec KEY  cipher.txt out.txt
Output written to: out.txt
#diff plain.txt out.txt
```

No output from diff ⇒ **encryption and decryption are correct**